The energy that becomes thermal

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for mechanical energy changes through heating and deformation.

SUCCESS CRITERIA - I can...

1. Track system boundaries in an energy balance.
2. Distinguish thermal energy from temperature.
3. Use energy differences to quantify a transfer.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “thermal energy” and “system” to explain your model or result. Sentence starters are optional.

thermal energy: internal energy associated with microscopic motion/interactions

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

A spring has k=300 N/m and stores 0.375 J.
Find displacement magnitude and force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $E_{mechanical,\ i} = E_{mechanical,\ f} + \Delta E_{thermal}$.
No other transfers in this model; include contacting materials in the system. Friction transfers mechanical energy to internal energy.

Worked reasoning: A 2 N friction force opposes a sliding box over 5 m. It does -10 J of work; in a suitable box-floor system this appears as a 10 J thermal-energy increase.

Your turn: Draw the force, displacement and energy transfer.
Name the system for your energy account.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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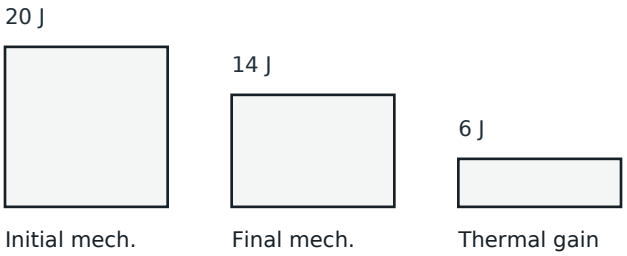
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A cart begins with 20 J mechanical energy and ends with 14 J. The closed accounting includes 6 J thermal increase. Does it balance?

Sentence starter: I notice … . This supports … because … .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: 1 2 3

04 Apply the physics

A 2 kg object slows from 5 to 2 m/s on a horizontal surface.
Find mechanical energy loss. If all becomes internal energy of object+surface, find that increase.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Does the same energy input always produce the same temperature rise in two different objects?
Name quantities you would need.

Sentence starter: Both … . They differ in … because … .

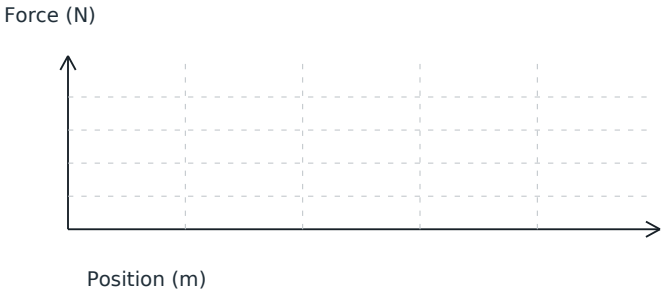
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Net force rises linearly from zero to 12 N over 0.4 m.
Find work and final speed for a 0.4 kg cart starting at rest.
Use the signed area to connect force and work.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 0.50 kg object at 6 m/s stops through friction.
Find energy transformed. If stopping distance is 1.5 m, find average opposing force.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 3 kg box slides down a 4 m ramp inclined 30° with friction magnitude 5 N. Starting at rest and using g=10, find final speed.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3